

Lab. 2 – Numerical and experimental analysis of T-junctions

Assigned: Friday April 10, 2026

Due: Friday April 24, 2026

Notes

To take the exam with the grading of submitted assignments + oral discussion it is necessary:

- to prepare **individually** the report of the lab, that must include the Matlab script, the obtained results and their brief comment;
- to submit the report **within the deadline**, uploading the PDF in the section “Elaborati” of the web portal

Consider the T-junction designed in problem 1 of the Assignment 2 and assume it is realized with microstrip technology.

The substrate is an FR4 material, with relative dielectric constant $\epsilon_r = 4.45$ and thickness $h = 1.55\text{mm}$. The central frequency is $f_0 = 5.8\text{ GHz}$.

A- Analysis of the T-junctions with CST.

- Calculate the length and width of the different lines (for the given values of the characteristic impedances) and implement the geometry in CST.
- Run the solver.
- Evaluate the frequency behavior of $|S_{11}(f)|$ and plot it as a function of f/f_0 in the range $[0.1f_0, 2f_0]$.
- Evaluate the output power ratio P_3/P_2 and plot it as a function of f/f_0 in the range $[0.1f_0, 2f_0]$.

B- Experimental characterization of the T-junction.

- Measure the frequency behavior of $|S_{11}(f)|$.
- From the measurement of the scattering parameters, evaluate the output power ratio P_3/P_2 and plot it as a function of f/f_0 in the range $[0.92f_0, 1.08f_0]$.

C- Compare the results related to $|S_{11}(f)|$ obtained with Matlab, CST and the experimental characterization of the device: export the data from CST and from the VNA, import them in Matlab™, and represent the three sets of data on the same plot, considering the range $[0.92f_0, 1.08f_0]$.

D- Compare the results related to power ratio P_3/P_2 obtained with CST and the experimental characterization of the device: export the data from CST and from the VNA, import them in Matlab™, and represent the two sets of data on the same plot, considering the range $[0.92f_0, 1.08f_0]$.